Principles of Synthetic Aperture Radar EO

Earth Observation Data Analysis

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SAPIENZA
UNIVERSITÀ DI ROMA



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Let's imagine ...

1 Chapter 1: The challenge

To be able to see trough the clouds



Surveillance.

Tracking.

Availability.

Let's imagine ...

1 Chapter 1: The challenge

To be able to see trough the dark



Surveillance.

Tracking.

Availability.

Let's imagine ...

1 Chapter 1: The challenge

To be able to see the Earth surface from hundreds of km



Synoptic view.

Frequent revisit time.

Monitoring.

Let’s imagine . . .

To be able to see small objects on the Earth surface from hundreds of km



Surveillance.

Monitoring.

Tracking.

Let's imagine ...

1 Chapter 1: The challenge

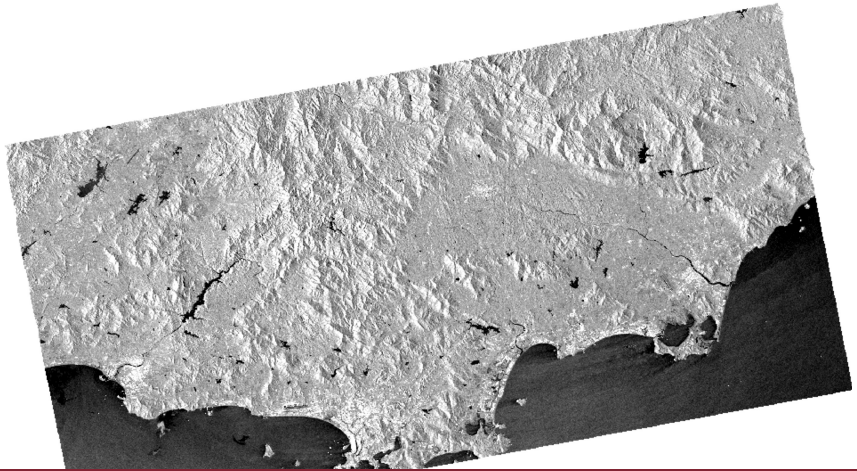
To be able to collect images of the observed area



Spatial information.

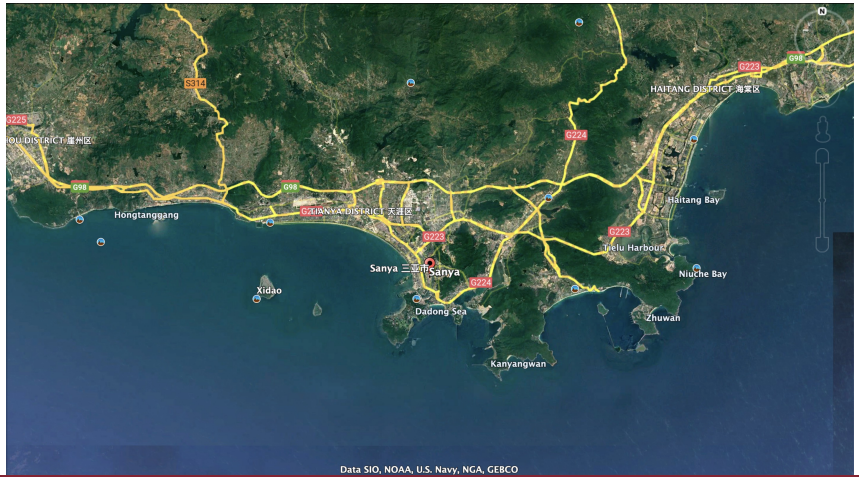
Just my imagination? No, the Synthetic Aperture Radar!

1 Chapter 1: The challenge



Sanya, Hainen, China

1 Chapter 1: The challenge

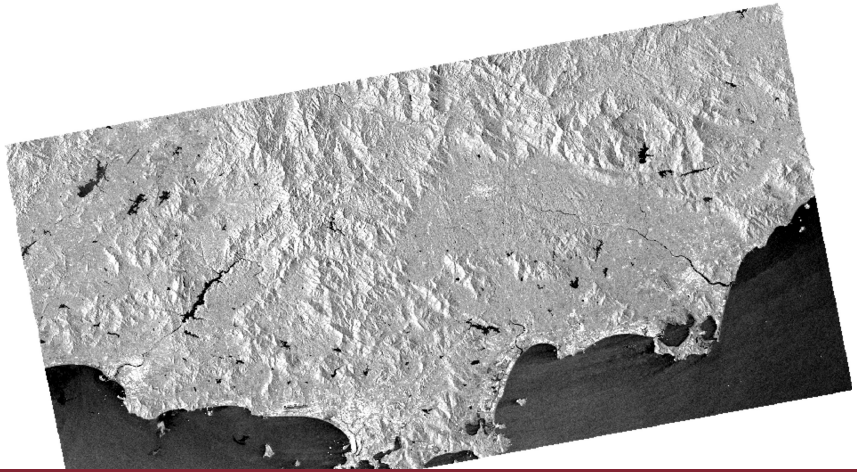


The optic scene - S2

1 Chapter 1: The challenge

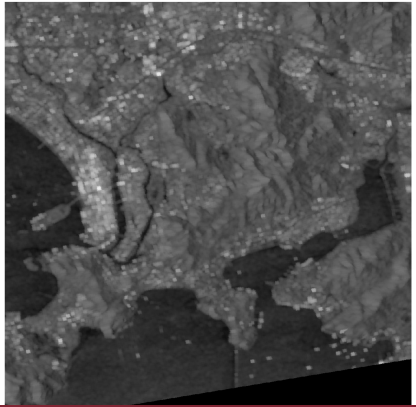


From a nice picture . . .



. . . look at the details. . . from hundreds of km. . . under cloudy conditions. . . during night-time





Colors. . . What does it mean?

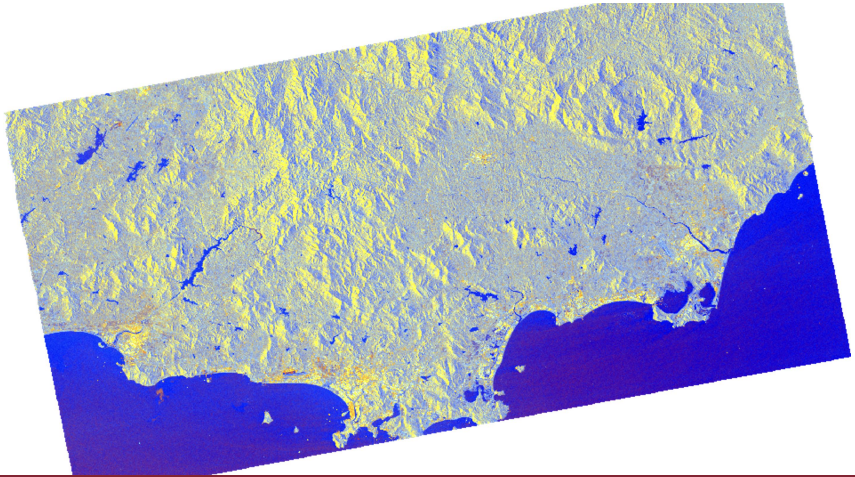




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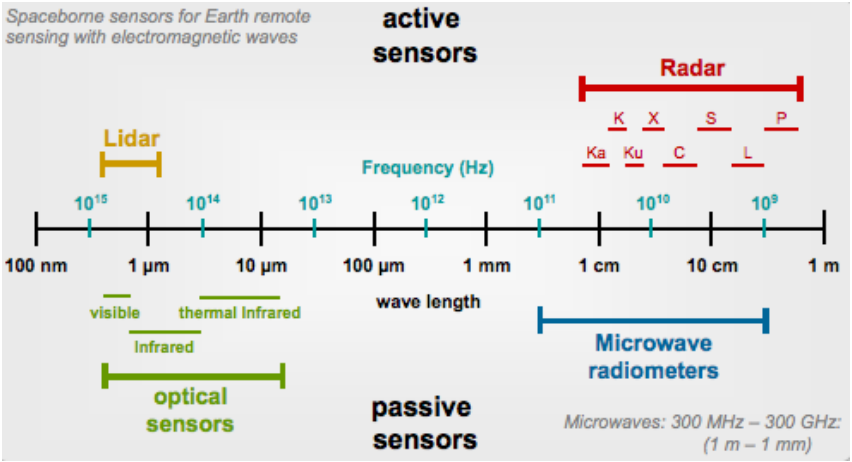
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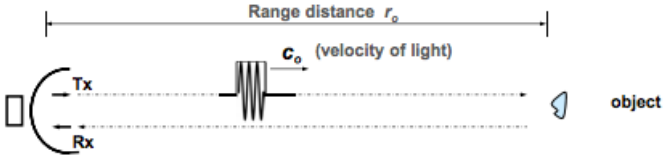
Once upon a time...

2 Chapter 2: The SLAR

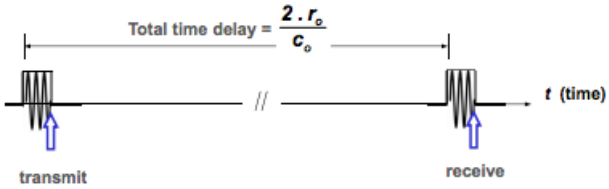


Microwaves


Spaceborne sensors for Earth remote sensing with electromagnetic waves
active sensors
Radar
K
X
S
P
Ka
Ku
C
L
Lidar
Frequency (Hz)
10^{15}
10^{14}
10^{13}
10^{12}
10^{11}
10^{10}
10^{9}
100 nm
1 µm
10 µm
100 µm
1 mm
1 cm
10 cm
1 m
visible
thermal Infrared
Infrared
wave length
optical sensors
Microwave radiometers
passive sensors
Microwaves: 300 MHz – 300 GHz: (1 m – 1 mm)



• Received echo signal (back-scattered signal of imaged object):



An active microwave sensor is a RADAR

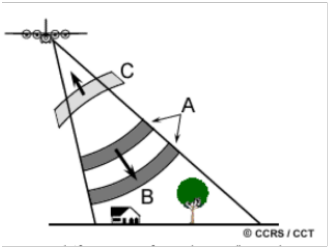
RADAR

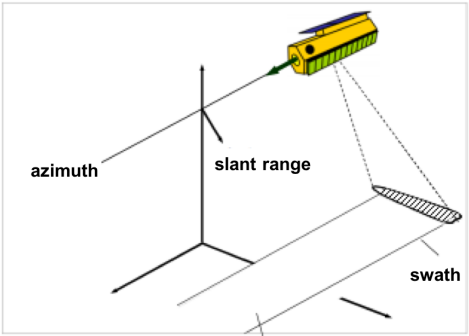
The sensor transmits a microwave (radio) signal towards the target and detects the backscattered portion of the signal. The strength of the backscattered signal is measured to discriminate between different targets and the time delay between the transmitted and reflected signals determines the distance (or range) to the target.

- RADARs provide their own source of microwave radiation to illuminate the target.

Imaging RADAR

2 Chapter 2: The SLAR



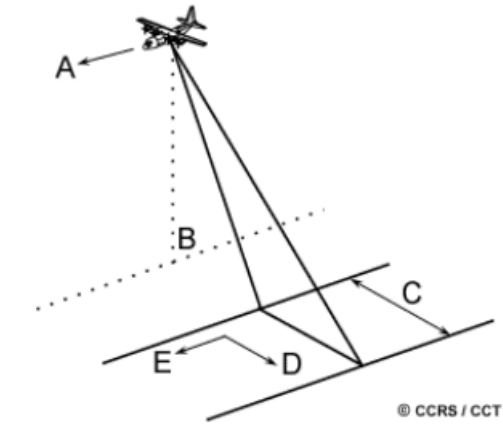




Imaging RADAR

- The transmitter generates successive pulses of microwave (A) at regular intervals which are focused by the antenna into a beam (B).
- The radar beam illuminates the surface obliquely at a right angle to the motion of the platform.
- The antenna receives a portion of the transmitted energy reflected (or backscattered) from various objects within the illuminated beam (C).
- The time delay between the transmission of a pulse and the reception of the backscattered “echo” from different targets, allow locating them.
- As the sensor platform moves forward, recording and processing of the backscattered signals builds up a two-dimensional image of the surface.

Side-looking view



The simplest radar system provides a 2-D reflectivity map of the imaged area, i.e., targets with high backscattered signal are identified as bright spots in the radar images and flat smooth surfaces as dark areas.



Side-looking view

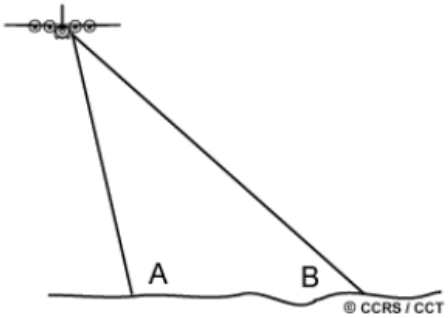
The imaging geometry of a radar system is different from the framing and scanning systems commonly employed for optical remote sensing.

- The platform travels forward in the flight direction (A) with the nadir (B) directly beneath the platform.
- The microwave beam is transmitted obliquely at right angles to the direction of flight illuminating a swath (C) which is offset from nadir.
- Range (D) refers to the across-track dimension perpendicular to the flight direction, while azimuth (E) refers to the along-track dimension parallel to the flight direction.

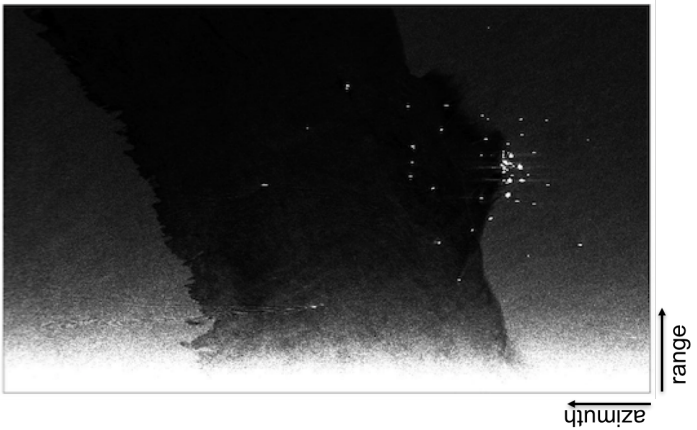
This side-looking viewing geometry is typical of imaging radar systems (airborne or spaceborne).

Side-looking view

The portion of the swath closest to the nadir track of the radar platform is called the near range (A) while the portion of the swath farthest from the nadir is called the far range (B).



From near range to far range

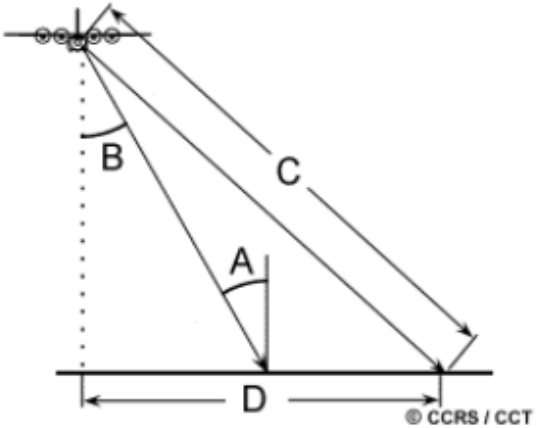




Side-looking view

- The incidence angle is the angle between the radar beam and ground surface (A) which increases, moving across the swath from near to far range.
- The look angle (B) is the angle at which the radar “looks” at the surface.
- At all ranges the radar antenna measures the radial line of sight distance between the radar and each target on the surface. This is the slant range distance (C).
- The ground range distance (D) is the true horizontal distance along the ground corresponding to each point measured in slant range.

Side-looking view





RAR spatial resolution

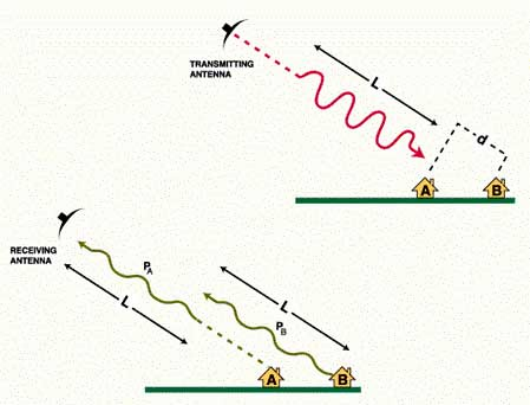
SLAR

An imaging RAR is the Side-Looking Airborne Radar (SLAR)

A single transmit pulse and the backscattered signal are used to form the image.

The spatial resolution is dependent on the effective length of the pulse in the slant range direction and on the width of the illumination in the azimuth direction.

Range resolution





Range resolution

For the radar to be able to distinguish two closely spaced elements, their echoes must necessarily be received at different times.

- The pulse of length L is approaching buildings A and B. The slant range distance between the two buildings is d.
- Since the radar pulse must travel two ways, the two buildings lead to two distinguished echoes if: $d > L/2$
- The part of the pulse backscattered by building A is PA , and the part of the pulse backscattered by building B is PB.
- It appears that to reach the target and come back, PB has covered an extra distance $2d$, and thus is at a slightly shorter distance than L behind PA.

The end of PA and the beginning of PB overlap when they reach the antenna. As a consequence, they are imaged as one single large target which extends from A to B!!!



Slant range resolution

The slant range resolution is given by:

$$\rho = \frac{c\tau}{2} \approx \frac{c}{2B} \quad ,$$

where c is the speed of light and τ is the pulse duration and B is the pulse bandwidth.

To improve range resolution, radar pulses should be as short as possible.

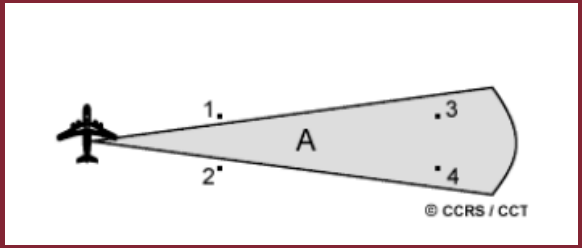
- Short pulses: low energy level to enable the detection of the reflected signals. Accordingly, its amplitude must be increased to keep the same total energy in the pulse.
- A high-energy pulse is difficult to build due to energy constraints.

CHIRP signal: instead of a short pulse with a constant frequency, a long pulse is emitted with a modulated frequency.

Azimuth resolution

Antenna diffraction limit

The azimuth or along-track resolution is determined by the angular width of the radiated microwave beam and the slant range distance.



This beamwidth (A) is a measure of the width of the illumination pattern. As the radar illumination propagates to increasing distance from the sensor, the azimuth resolution increases (becomes coarser).



Azimuth resolution

Two targets in the azimuth or along-track resolution can be separated only if the distance between them is larger than the radar beamwidth.

- The radar beamwidth is: $W = \frac{\lambda}{L}$. Where L is the along-track antenna length and λ is the em wavelength.

Accordingly, the azimuth resolution is:

$$\Delta_x = R * W = \frac{R\lambda}{L} \quad ,$$

where, R is the slant range distance.

A longer antenna (or aperture) will produce a narrower beam and finer resolution.



Technical challenges to be addressed

Range spatial resolution

High energy level pulses.

They are very energy-consuming and not suitable especially for satellite platforms where the energy source comes from solar panels.

Azimuth spatial resolution

It depends on the antenna physical size and on the range.

Not suitable for satellite platforms since it calls for very large antennas to get fine spatial resolution.



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Azimuth resolution

SAR processing

SAR acquisition modes

Speckle

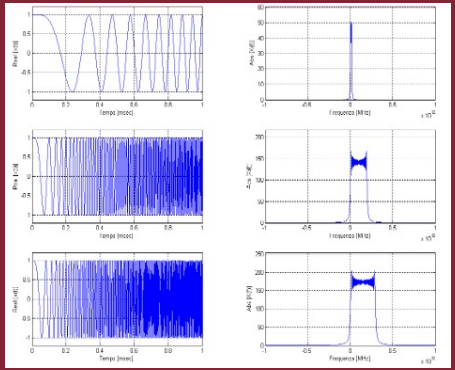
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A chirp signal is transmitted



A longer linearly modulated pulse is transmitted. The energy of this pulse is distributed over a longer duration, but it can be compressed again after receiving it by a *matched filtering operation*.

TX and RX chirp signals

A chirp signal is transmitted



$$f_t(t) = e^{j\left(\omega t + \frac{\alpha t^2}{2}\right)} rect\left(\frac{t}{\tau}\right)$$

j is the imaginary unit;

ω is the angular frequency;

α is the chirp rate;

τ is the pulse duration.

TX and RX chirp signals

An attenuated and retarded chirp is received



$$f_r\left(t-\frac{2r}{c}\right)=e^{j\left[\omega\left(t-\frac{2r}{c}\right)+\frac{\alpha}{2}\left(t-\frac{2r}{c}\right)^2\right]}rect\left(\frac{t-r/2c}{\tau}\right)$$

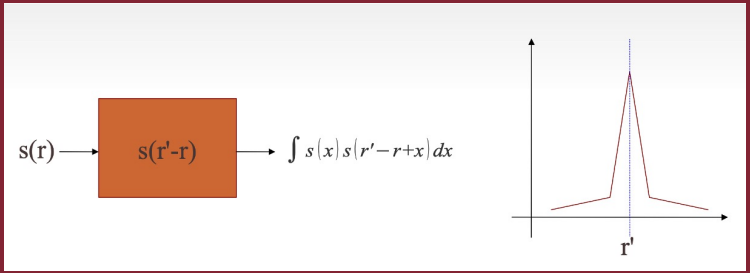
a point target is assumed;

can be easily generalized to distributed scene calling for the reflectivity $\gamma(\cdot)$;

To improve the spatial resolution pulse compression is needed!

TX and RX chirp signals

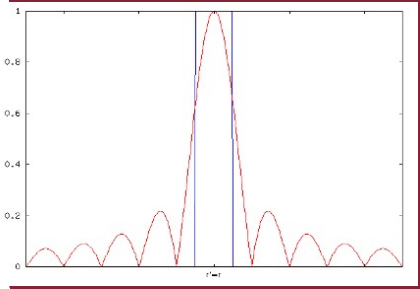
A matched filter is built...



that contrasts the RX chirp with a replica of the TX one in order to perform pulse compression

TX and RX chirp signals

The compressed signal reads as follows:



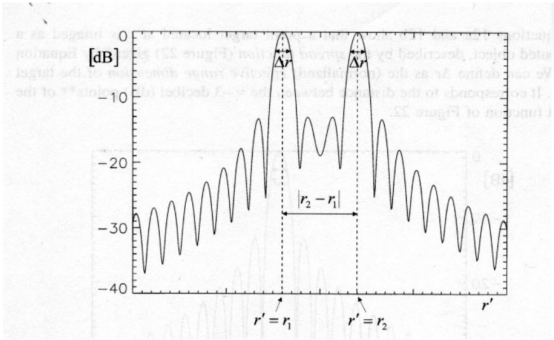
$$f_{out}(r') = e^{-j\frac{4\pi}{\lambda}r} sinc\left(\frac{\pi}{\Delta r}(r' - r)\right)$$

$\Delta r = \frac{c}{2B}$

A point target located at r is imaged as a distributed target Δr

To improve the spatial resolution pulse compression is needed!

Range resolution



Adopted from Franceschetti et al. 1999

$$|r_2 - r_1| \geq \Delta r$$

Fine range spatial resolutions can be achieved using large pulse duration coupled with large chirp rates

From the real aperture...

Real Aperture Radars do not provide fine azimuth resolution in space-borne configuration, while they can be successfully used when dealing with airborne configurations.

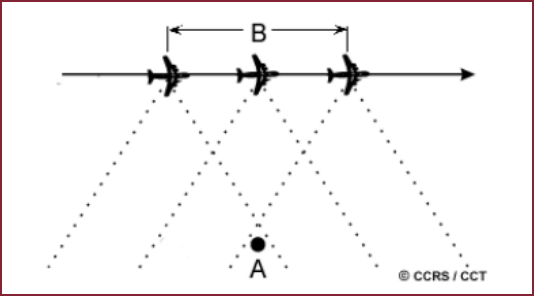
...to the synthetic aperture

To overcome this limitation, the forward motion of the platform and special recording and processing of the backscattered echoes are used to simulate a very long antenna and thus increase azimuth resolution.

SAR azimuth resolution

Synthetic array

SAR systems achieve good azimuth resolution that is independent of the slant range to the target, yet use small antennae and relatively long wavelengths to do it.

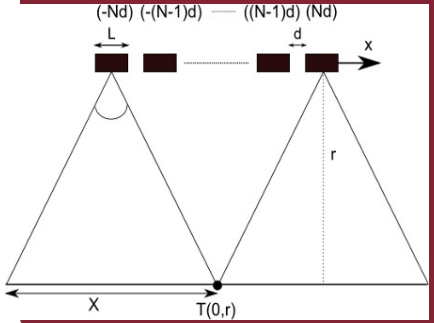




Synthetic array concept

- As a target (A) first enters the radar beam (1), the backscattered echoes from each transmitted pulse begin to be recorded.
- As the platform continues to move forward, all echoes from the target for each pulse are recorded during the entire time that the target is within the beam.
- The point at which the target leaves the view of the radar beam (2) some time later, determines the length of the synthesized antenna (B).
- This method of achieving uniform, fine azimuth resolution across the entire imaging swath is called SAR.
- The azimuth spatial resolution is $\frac{L}{2}$.

Consider 2N+1 acquisitions equally spaced (d), obtained by moving the real antenna along with the flight direction



The echo received by the n-th position reads:

$$f_r(n'd) = e^{-j\omega\frac{2R}{c}} \approx e^{-j\omega\frac{2r}{c}-j\frac{2\pi}{\lambda r}(n'd)^2}$$

$$n' = -N \dots N$$

$$R = \sqrt{r^2 + (n'd)^2} \approx r + \frac{(n'd)^2}{2r}$$

$$X \approx r\frac{\lambda}{r}$$

The signal received in the azimuth channel

The second exponential in the received signal is similar to the one measured in the range channel:

$$f_r(n'd) \approx e^{-j\omega\frac{2r}{c}-j\frac{2\pi}{\lambda r}(n'd)^2}$$

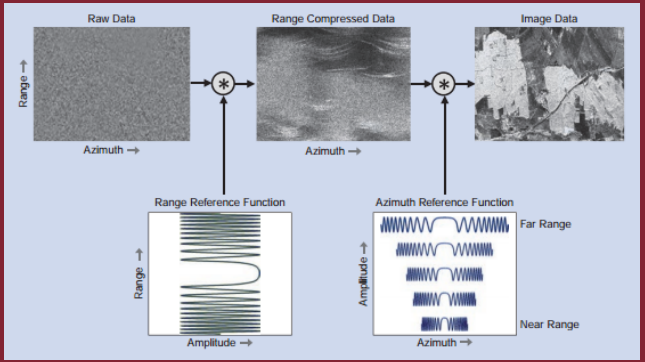
This implies that a convolution - similar to the one of the range channel - can be performed. In the neighbors of the point target the spreading function reads:

$$f_{out}(x') \approx sinc\left(\frac{\pi}{\Delta x}x'\right)$$

with $x' = \frac{n'd}{X}$

$\Delta x = \frac{L}{2}$ under the hypothesis that $d \leq \frac{L}{2}$

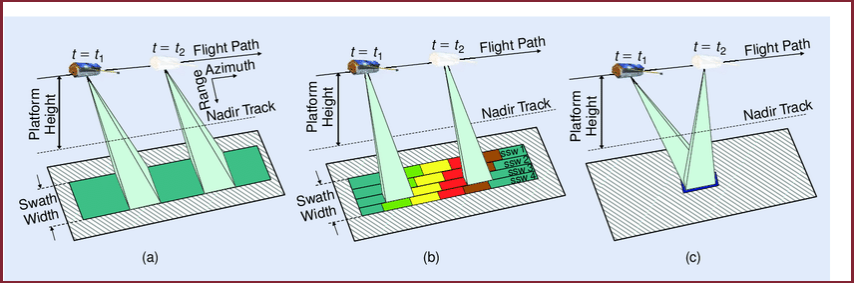
In case of distributed targets



SAR processing can be efficiently performed in the Fourier domain

Trading off swath vs spatial resolution

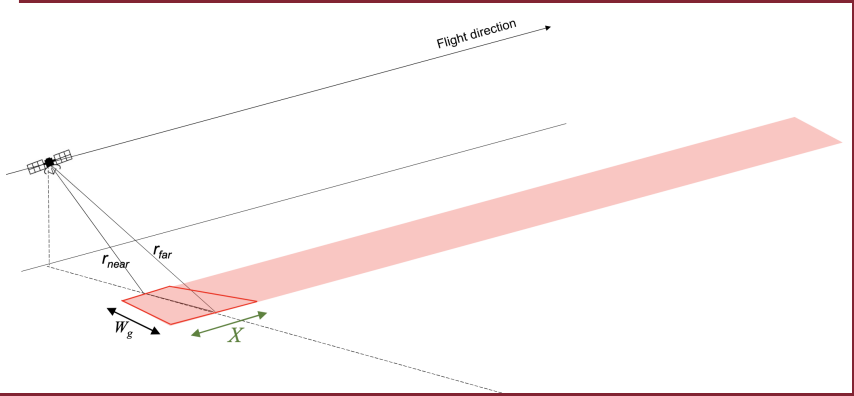
Three main imaging modes



Thanks to the SAR antenna that consists of sub-apertures whose amplitude and phase can be controlled to “steer” the antenna pattern

StripMAP

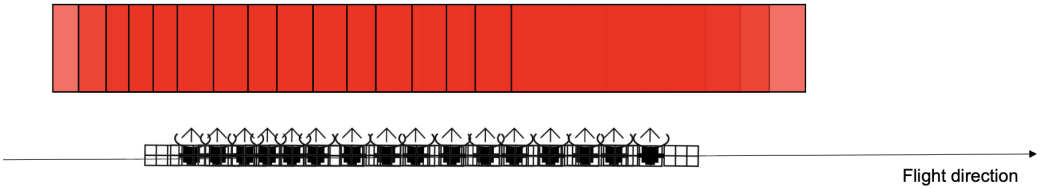
The pattern is fixed to one swath, thus imaging a single continuous strip



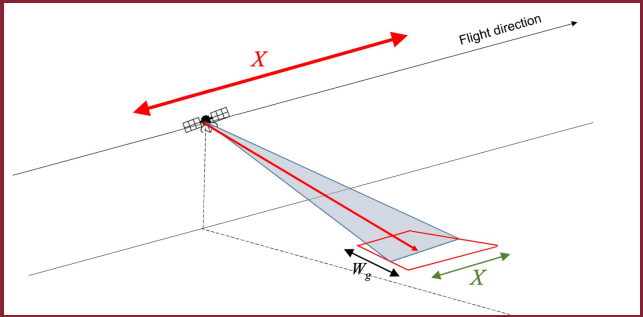
Good compromise between swath width and spatial resolution

StripMAP

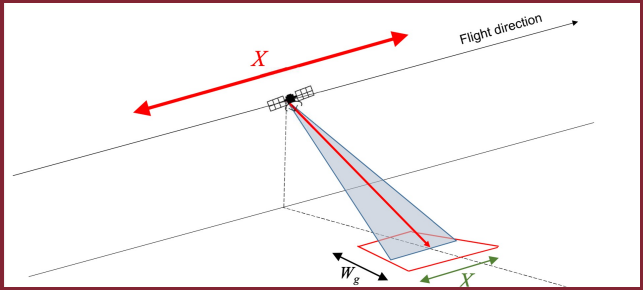
3 Chapter 3: The Synthetic Aperture Radar (SAR)



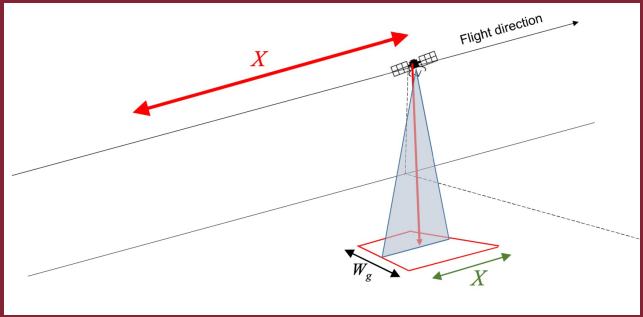
The pattern is steered in azimuth towards a fixed point to illuminate a given region



The pattern is steered in azimuth towards a fixed point to illuminate a given region

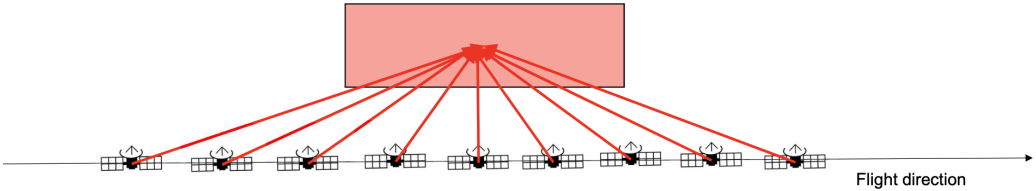


The pattern is steered in azimuth towards a fixed point to illuminate a given region



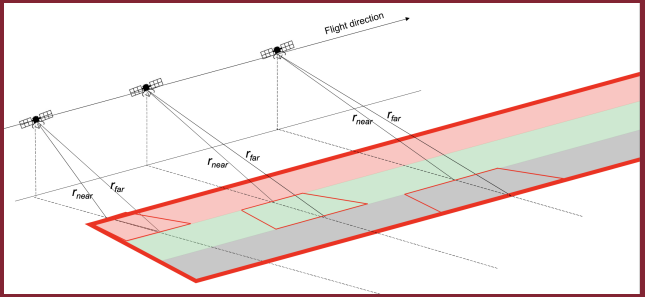
Spotlight

3 Chapter 3: The Synthetic Aperture Radar (SAR)

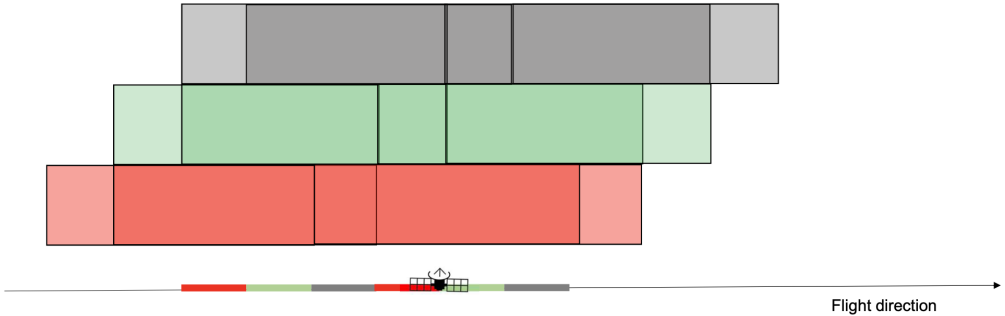


Fine spatial resolution coarse swath

The elevation pattern is successively steered to different elevation angles corresponding to multiple sub-swaths



ScanSAR



Coarse spatial resolution very large swath

Acquisition modes: in a nutshell

Stripmap

No beam steering

No burst mode

Spotlight

Azimuth beam steering

No burst mode

Azimuth resolution

Azimuth coverage

ScanSAR

Range beam steering

Burst mode

Range coverage

Azimuth resolution; scalloping effect

TOPSAR

Azimuth beam steering

Range beam steering

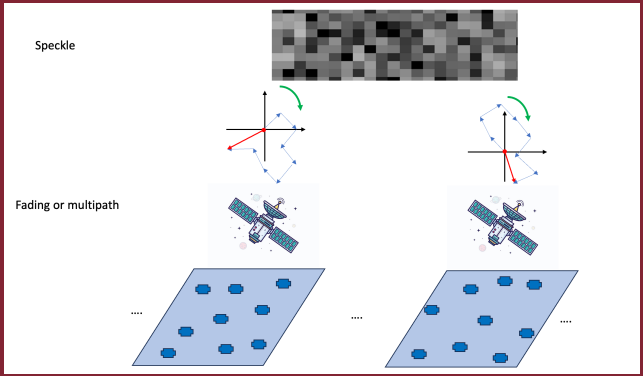
Burst mode

Range coverage; scalloping mitigation

Azimuth resolution

Speckled SAR imagery

The SAR resolution cell consists of a number of randomly distributed elementary scatterers



Speckled SAR imagery

From fading to speckle

Two generic resolution cells call for a different distribution of elementary scatterers

The signal resulting from the coherent combination of the scatterers' contributions may change significantly from a cell to another.

Those fluctuations are termed as fading.

This leads to fluctuations (in both intensity and phase) on a pixel basis.

Those fluctuations in the SAR image plane are termed as speckle.

The speckle is the fading measured by the SAR system



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Polarimeter

A device that measures the polarization of electromagnetic waves is called polarimeter



Microwave polarimeters

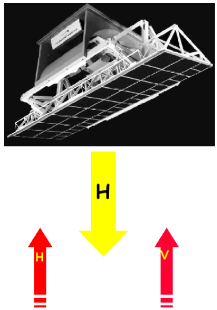
Radar

Radiometer

Polarimetric SAR measurements

PolSAR transmits a linear polarization (H/V) and measures both the co- and the cross-polarized scattered waves (quad-pol).

$$\mathbf{E}^s = \mathbf{S}\mathbf{E}^i$$

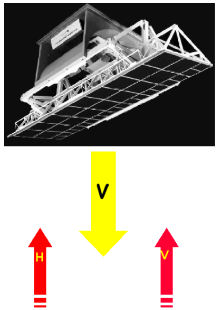


$$\begin{pmatrix} S_{hh} & S_{hv} \end{pmatrix}$$

Polarimetric measurements

PolSAR transmits a linear polarization (H/V) and measures both the co- and the cross-polarized scattered waves (quad-pol).

$$\mathbf{E}^s = \mathbf{S}\mathbf{E}^i$$



$$\begin{pmatrix} S_{hh} & S_{hv} \\ S_{vh} & S_{vv} \end{pmatrix}$$

S is termed as scattering matrix

Each complex $\mathbf{S}$ element is termed as scattering amplitude.

In the backscatter case and under reciprocity assumption $S_{hv} = S_{vh}$.

It is a first-order descriptor.

To effectively extract polarimetric scattering information related to the observed scene, second-order descriptors are needed.



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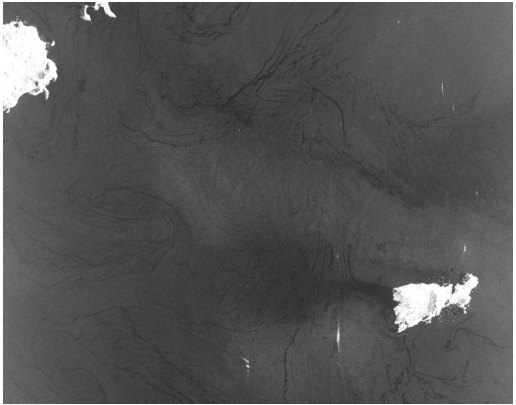
PolSAR measurements

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A physical understanding

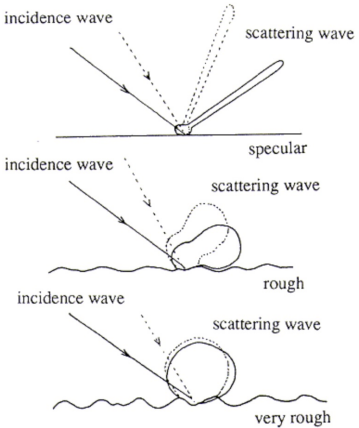
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A physical understanding of microwave sea surface scattering is a mandatory step to correctly interpret SAR imagery and to generate added-value products

Sea surface scattering

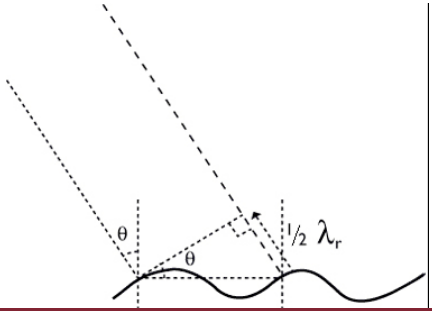
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The sea surface roughness scales rule the behavior of the signal scattered back to the receiving antenna

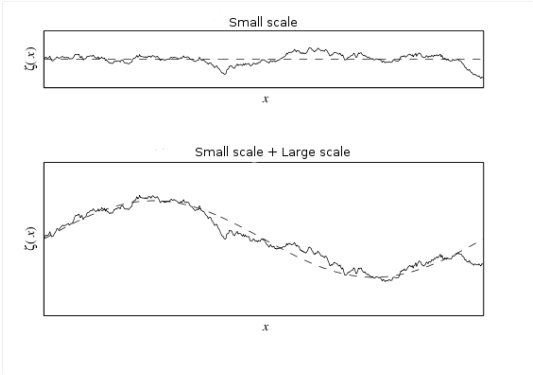
Bragg scattering

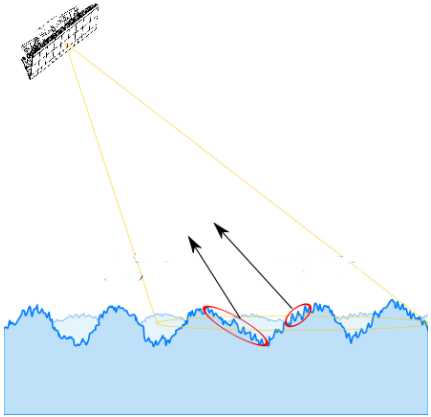
The scattering results from the Fourier components of the surface spectrum producing a constructive interference in the scattering direction



Sea surface scattering

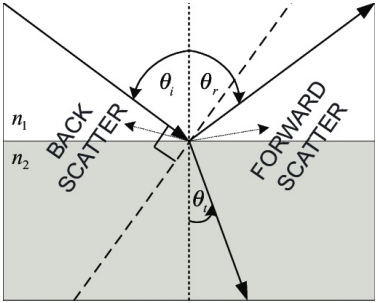
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Microwave scattering…is important

Different technology, different applications…the same theoretical structure!



- Monostatic configuration.
- Bistatic configuration.
- Multi-static configuration.



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For further reading

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- A. Moreira et al. “A Tutorial on Synthetic Aperture Radar,” *IEEE Geoscience and Remote Sensing Magazine*, March 2013. Availbe at: http://www2.geog.ucl.ac.uk/ mdisney/teaching/PPRS/PPRS_7/esa_sar_tutorial.pdf
- G.Franceschetti and R.Lanari, Synthetic Aperture RADAR processing, CRC Press LLC, United States, 1999.
- Canada Centre for Remote Sensing (CCRS), Glossary of Remote Sensing, available at: `http://ccrs.nrcan.gc.ca/glossary/index_e.php`